

Edouard Murat

Gameplay Programmer & Team Lead

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French: native • English: fluent

PROFILE

Gameplay programmer and team lead with 20 years of software experience, including five years of production C#. Led a six-person Godot team building curriculum-aligned educational games for Android. Specializes in gameplay systems, AI behavior, procedural generation, shader programming, and reusable game frameworks. Five games live on Steam in a free public demo, with deployment pipelines for Google Play and the Apple Store.

TECHNICAL SKILLS

Game Engines	Godot, Unity
Languages	GScript, C#, Python, Java, HTML/CSS/JS, SQL
Shaders & Graphics	Shader scripting (Godot, Unity)
Gameplay Systems	Input (keyboard & mobile), AI behaviors, procedural generation, physics-based mechanics, state machines, combat systems
Tools & Frameworks	Git, GitHub, Blender, VS Code, Django, SQLite, Audacity, Linux
Other	Agile workflows, UI integration, performance optimization

EXPERIENCE

Lead Gameplay Programmer (Godot) — Sigma Digital, South Africa (remote) Sept 2025 – Aug 2026

Led a six-person Godot team building curriculum-aligned literacy games delivered on Android tablets to first- and second-grade children in Africa.

- Hired the programming team and led it as technical lead, owning architecture, code review, and sprint planning
- Designed the platform's UI framework and an MVVM + FSM architecture adopted across all activities, decoupling game logic from presentation, enabling per-activity unit testing, and making curriculum content swappable without rebuilding core mechanics
- Delivered six mini-games and supported two rounds of playtesting with the target age group
- Built for low-cost Android tablets, with gameplay mapped to a first- and second-grade reading and writing curriculum
- Worked directly with educators and instructional designers to translate curriculum goals into game mechanics, bridging educational theory and interactive design across each sprint
- Provisioned GPU-enabled AWS EC2 instances to run automated Godot UI tests, enabling headless rendering for CI without local hardware

Gameplay Programmer (Godot) — Shattered Paradise, One More Turn Studios (remote) Jul 2025 – Dec 2025

Sole programmer on a turn-based grid-based dungeon crawler roguelike (Dante's Inferno theme, procedurally generated dungeons). The creative director defined the vision; owned all technical work end-to-end.

- Wrote all Godot gameplay code, shaders, the game website, and backend from scratch
- Prioritized and built features independently across the full stack

Gameplay Programmer (Godot) — Pursuit of Life, Dreamboat (remote) Jun 2025 – Aug 2025

Shader-focused contract on a pixel-art urban life-sim.

- Built a dynamic rain system with customizable impact zones and per-region wetness tracking
- Implemented real-time puddle reflections for sprites, NPCs, and swaying foliage
- Developed day/night lighting driven by a global time-of-day system

Game Developer — Playworks North, Montreal 2022 – Present

Founder and lead developer of an educational gaming platform blending custom-built quizzes with gameplay across farming, hack & slash, platformer, puzzle, and sports genres.

- Shipped five games in Unity and Godot, live on Steam as a free public demo since September 2026, integrating real-time quizzes into diverse gameplay loops
- Designed and implemented gameplay systems: player control, AI, procedural generation, level logic, scoring, and feedback
- Implemented unified input systems supporting both keyboard and mobile touch controls
- Created custom shaders for visual storytelling and UX (X-ray vision, mini-cameras, 3D text, underwater effects)
- Built a full-stack Django backend: authentication, data storage, in-game unlockables, and content delivery
- Built and maintained deployment pipelines for Steam, Google Play, and the Apple Store; took the Steam demo through Valve review to public release

Web Developer and Graphic Designer — HeadPause, Montreal	2018 – 2020
- Developed and maintained the company website and online presence - Created animations and visual assets, and ran social advertising campaigns	
Java and Web Developer — BMC Software, Montreal	2013 – 2014
- Partial management of the development team in SCRUM, including code reviews and TDD - Built a new adaptive website to client requirements alongside parallel remote teams - Interviewed candidates for web developer positions	
C# Developer and Analyst — Fujitsu Consulting, Montreal	2008 – 2013
- Five years of production C#: a payment solution for HydroSherbrooke and the daycare management system for the Ministère de la Famille du Québec - Analyzed client needs, resolved blocking issues, and trained new developers across a three-year project	

SELECTED GAME PROJECTS

Full list of playable games at edouardmurat.com

Base of 30 (Godot) — Rogue hack & slash through procedurally generated rooms	2025
- Designed and implemented all core gameplay systems, including combat and player progression - Developed procedural generation for room layouts, obstacle placement, and navigation paths - Built custom behavior trees for enemy types with unique attack patterns and movement logic	
Aerial Ambush (Godot) — Skill-based platformer with homing missiles	2025
- Implemented responsive, high-speed player movement with precise collision handling - Developed homing missile AI with adjustable targeting and physics-based steering - Designed and balanced dodge-based gameplay loops for short sessions and high replayability	
Agent Ka (Godot) — Waldo-like game in a generated city with crowds and powers	2024
- Built a procedural city generation system with randomized buildings, roads, and crowds - Programmed crowd simulation with dynamic movement, behaviors, and density-based logic - Implemented X-ray, mini-camera, and slow-motion powers via custom shaders	
Veggies Island (Godot) — Farming game on a cartoonish island	2024
- Implemented the core farming loop: planting, watering, harvesting, and crop growth stages - Built procedural island generation with randomized terrain, resources, and farm zones - Developed a lightweight inventory system for harvested crops and seeds	
Pico Golf (Godot) — Mini golf challenges with physics-based terrain	2023
- Designed 3D mini-golf courses with increasing difficulty and varied sea-floor layouts - Integrated animated fish and custom water and caustics shaders to enhance immersion - Developed audio-visual feedback systems for a calming, low-pressure experience	
Tomb (Godot) — Doom-inspired shooter for Godot Wild Jam #42	2022
- Built from scratch in under one week during a jam, demonstrating rapid prototyping - Engineered arcade-style shooter feel: player movement, shooting, and health systems - Designed enemy AI using state machines and pathfinding logic with trap systems	

Earlier titles include Left 4 Dead 2D, Meditation Card Game, Fruit, Archer, and Meditation World VR.

EDUCATION

B.A. Psychology — Concordia University, Montreal	2015 – 2017
B.Eng. Computer Engineering — University of Sherbrooke	2005 – 2009
College Diploma, DEC in Sciences — Stanislas College, Montreal	2003 – 2005

CERTIFICATIONS

Certified Associate Python Programmer — Python Institute	2024
Oracle Professional Certification, Java SE 7 Programmer	2013
Oracle Associate Certification, Java SE 7 Programmer	2012